COHORT 16 DSML- MINI PROJECT DOCUMENTATION

Renewable Energy Transition Speed

Date:02/03/2026

DAY 5: MODEL EVALUATION

Objective

- Evaluate the trained models (Linear Regression and XGBoost).
- Identify the best-performing model for predicting **Renewable Energy Transition Speed (SDG-7)**.

Work Done

Evaluation Setup

- Loaded the trained Linear Regression model.
- Used the unseen test dataset for unbiased evaluation.
- Generated predicted values for transition speed.
- Compared predicted values with actual values.
- Implemented standard regression evaluation metrics (MAE, RMSE, R^2).

Evaluation Metrics

- **Mean Absolute Error (MAE):** Measures average absolute prediction error; easy to interpret.
- **Root Mean Squared Error (RMSE):** Penalizes larger errors; indicates stability.
- **R^2 Score:** Measures proportion of variance explained; indicates goodness of fit.

Performance Observations

- Linear Regression captured the overall trend in renewable transition speed.
- Produced stable predictions on the test dataset.
- Minor deviations due to possible nonlinear relationships in energy data.
- No significant overfitting detected.

Outcome :

Linear Regression was finalized as the project model due to its stable performance, high interpretability, and suitability for the mini-project.

Group Discussion : Day 5

Figure: Team Meeting – 02nd March 2026